

Analysis of Flight Risk, Delays, and Cost Efficiency

by

Andrew Lee

Adam Mills

Dat Nguyen

Mihika Shah

Sophia Kreiler

Jacqueline Le

Introduction

Jeju Air international flight 7C2216 tragically collided with an embankment on Dec. 29, 2024, marking it as the deadliest aviation disaster in South Korea.¹ With multiple accidents making headlines, consumer faith in airline companies has declined sharply over the past year. Concerns over aircraft reliability and risk management continue to trouble passengers, driving the need for research into recent data trends to assess the safety and efficiency of air travel.

This project aims to analyze flight data and develop risk profiles for routes and airlines by examining costs, delay probabilities, severity, and reliability. Specifically, the study evaluates the likelihood of delays for individual airlines and examines the severity of delays when they occur. It also analyzes cost-per-mile metrics for each airline and explores the relationship between delay frequency, severity, and cost efficiency. Furthermore, the project identifies routes that exhibit higher-than-average delays and determines which airlines are most susceptible to these delays.

By addressing inefficiencies and identifying high-risk factors in the airline industry, the project seeks to improve operational performance, financial outcomes, and passenger satisfaction. With a focus on commercial airlines, this research is both relevant to current industry challenges and broad enough to ensure meaningful insights.

Literature Review

Similar research conducted on flight data has been studied over time to analyze aviation trends, flight crashes, comparisons of plane manufacturers, and more. The *Human Error and Commercial Aviation Accidents: A Comprehensive, Fine-Grained Analysis Using HFACS*, researched by individuals across various institutions through the Federal Aviation Administration, studied factors such as aircrew, environmental, supervisory, and organizational influences that have resulted in aviation accidents. Commercial aviation accident data from 1990 to 2002 was used in their method of analysis. The authors used HFACS as the main analytical tool and performed a descriptive statistical analysis on accident reports, coding each one according to the HFACS categories. Additionally, a comparative cross-tabulation was performed

¹ Kawoosa, Vijdan Mohammad, Han Huang, Rebecca Pazos, and Simon Scarf. "The Final Moments before the Jeju Air Crash in South Korea." Reuters, December 30, 2024.

to examine the relationship between error types and accident outcomes. This study can be compared to our data as both identify risk factors that contribute to operational failures. Both research projects include data-driven decision-making in aviation.

Analysis and Prevention of MCAS-Induced Crashes examines the failures of the Maneuvering Characteristics Augmentation System (MCAS) in the Boeing 737-Max aircraft. The methodology used in the MCAS crash analysis by Curran et al. (2024) and the flight delay and cost-per-mile prediction project both rely on modeling and data-driven techniques but differ significantly in purpose and implementation. The MCAS study focuses on safety-critical control systems, using simulation environments like MATLAB/Simulink and JSBSim to model aircraft behavior under sensor faults and pilot error, ultimately proposing a semi-autonomous system (SA-MCAS) that dynamically balances control between automation and human input.

In contrast, our project uses supervised machine learning models—Support Vector Machines (SVM) and Multi-Layer Perceptrons (MLP)—to predict flight delay severity (classification) and cost-per-mile (regression), emphasizing performance metrics like accuracy, precision, recall, F1-score, RMSE, and R^2 . While the MCAS approach prioritizes real-time robustness and interpretability in safety contexts, our project focuses on predictive accuracy, generalization (via validation loss and early stopping), and post-hoc error analysis for operational insights.

The article *Competition in the Cockpit* examines financial pressures and disruptions affecting airline reliability, which is crucial for understanding the broader context of cost efficiency, an aspect that aligns with our project's goal of predicting cost-per-mile using machine learning models. While our project uses a quantitative, data-driven methodology—employing SVM and MLP models with metrics like accuracy, precision, recall, RMSE, and R^2 —the article relies on a qualitative approach, using case studies (like the Boeing 737 MAX crashes) and secondary data sources such as financial records, regulatory reports, and expert commentary. Its focus is on policy analysis and economic forces, evaluating how deregulation and cost-cutting measures have contributed to operational instability in the airline industry. In contrast, our model-based approach provides flight-specific predictive insights. Together, both perspectives

offer complementary understandings: one from a strategic, structural viewpoint, and the other from an operational, predictive lens.

Our project shifts the focus from accident prevention to operational performance prediction, using machine learning techniques such as SVM and MLP to anticipate flight delay severity and cost-per-mile. While the HFACS and MCAS studies examine aviation failures post hoc to inform systemic improvements, our work is rooted in proactive risk profiling, helping airlines identify inefficient or high-risk routes before they impact passengers or finances.

Dataset

The dataset used for this project, *flight_data_2018_2024.csv*, contains detailed information on flights over six years. It includes features such as departure and arrival delays, distance, elapsed times, and categorical details like airline, origin, and destination, capturing various factors that influence flight performance. After preprocessing, the dataset was cleaned by removing canceled and diverted flights, handling missing values through forward filling, and encoding categorical variables using one-hot encoding for compatibility with machine learning models.

The cleaned dataset, *cleaned_flight_data.csv*, consists of 14 core features, including numerical fields like Distance and AirTime, as well as engineered features such as CostPerMile (a metric calculated from elapsed time and distance) and DelaySeverity (a categorical representation of delay severity). These features are designed to support both regression and classification tasks. For binary classification, the DelayBinary column was created, indicating whether a flight was delayed or on time.

Baseline

The baseline models were designed to establish a standard for evaluating the performance of the methods. For delay severity prediction, a static model was implemented using predefined thresholds, categorizing flights with delays of 15 minutes or less as on time and those exceeding 15 minutes as delayed. This approach established initial metrics for assessing classification performance. Similarly, for the cost-per-mile prediction task, a regression baseline model based on linear regression was utilized, using features such as Distance and CRSElapsedTime to

estimate cost-per-mile. These baseline models offered dependable benchmarks for comparing different approaches, like SVM, MLP classifiers, and MLP regressors in both classification and regression tasks.

Main Approach

To predict delay severity and cost-per-mile, advanced machine learning techniques, including SVM and MLP, were utilized to deliver actionable insights into airline operations. For delay severity classification, SVM leveraged features like distance, departure hour, and one-hot encoded locations, using an RBF kernel to capture non-linear patterns. In parallel, the MLP classifier employed a neural network with dropout layers to prevent overfitting and EarlyStopping to optimize training efficiency. Evaluation metrics, including precision, recall, F1-score, and confusion matrices, demonstrated that while SVM performed well on simpler patterns, the MLP classifier excelled in identifying more complex feature interactions, albeit with longer training times.

For cost-per-mile regression tasks, scikit-learn's MLPRegressor and a customizable Keras-based MLP were employed. The scikit-learn model, featuring two hidden layers and normalized numerical features, minimized prediction errors using MSE loss and was assessed with RMSE and R^2 scores. The Keras-based regressor offered greater flexibility with a larger network architecture, dropout layers, and the Adam optimizer, achieving improved performance and generalization over baseline models. Visualizations comparing actual versus predicted values highlighted strong accuracy with minimal outliers, enhancing trust in the predictions.

Evaluation Metric

Our project involves two different learning tasks: a classification task to predict flight delay severity and a regression task to predict cost per mile. Each task uses appropriate evaluation measures.

For the classification task, we use metrics such as accuracy, precision, recall, and F1-score in a confusion matrix. Accuracy measures the proportion of correct predictions, the fraction of correctly classified flights, while precision and recall focus on the positive class (delayed flights). Precision measures the proportion of flights that were predicted as delayed and are delayed to penalize false alarms. Recall measures the proportion of actual delayed flights

correctly identified to ensure the most delayed flights are caught. Last but not least, F1-score is a blend between precision and recall to balance false positives and false negatives. With a confusion matrix, we can visualize the breakdown of predictions versus actual classes to highlight which classes are most often confused (e.g., on moderate delay vs severe delay). This can help identify the systematic biases.

For the regression task, we use error metrics that quantify prediction deviations and variance, such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2). Lower RMSE means predictions are close to the true costs, while MAE gives the average absolute difference (treating all errors equally). Together, RMSE and MAE give us a sense of prediction accuracy. R^2 shows us how well the model explains the variance in cost-per-mile. Low RMSE values indicated accurate predictions, while encouraging R^2 scores showed that the models captured most of the variance in cost-per-mile.

During model training, we also observe validation loss to detect overfitting. We used a 20% validation split from training data and applied early stopping for neural networks when the validation loss stops improving to help ensure the model generalizes well. Finally, we perform error analysis and visual diagnostics by plotting actual vs predicted values to provide qualitative insight. Together, these metrics and tools ensure reliable models that support better airline operations.

Results and Analysis

The results demonstrate the performance of our models for both binary classification and regression tasks, highlighting key findings and areas for improvement. The models effectively detected delayed flights, with high recall ensuring most delays were identified, though lower precision indicated some false positives.

Delay Prediction:

For classifying and predicting delays, we implemented both SVM and MLP models. Our main classifier is the SVM model with an RBF kernel. We sample 25,000 flights per major airline to manage data volume and use an 80/20 split and class balancing to address label imbalance. On this dataset, the SVM model achieves a high overall accuracy (about 85-92% across airlines). For example, United Airlines (“UA”) achieves ~86% accuracy, and Southwest

(“WN”) flights reach ~92% accuracy. The SVM also attains a very high recall on delayed classes (~86%), which means that it correctly identifies most delayed flights. However, precision is lower on delay classes, indicating some false positives. The precision for “Moderate” delay is around 0.53-0.72, and “Severe” delay is around 0.68-0.86. It means that the model tends to overpredict delays when optimizing for recall. In all cases, the “On-Time” class has both high precision and recall since most flights are on time.

The other classifier for our delay prediction was an MLP model with two hidden layers (128 and 64 units) and dropout rates of 0.3 and 0.2, respectively. It employed the Adam optimizer and binary cross-entropy loss. The MLP achieves about 60% accuracy, but 0% recall on the delayed class because it never flagged a flight as delayed. The SVM implementation outperformed the MLP as reflected in the improved accuracy, recall, and precision metrics. These results mean that the model can both identify the delayed class and classify it correctly using SVM. On the other hand, the MLP implementation struggled to identify the delayed cases and classify them correctly.

Aspect	SVM	MLP
Accuracy	88%	60%
Recall (Delayed)	High (86.6%)	Average (50%)
Precision (Delayed)	Strong (78.6%)	Low (30%)
Model Flexibility	Linear, fixed kernel	Deep, non-linear adaptable
Imbalance Handling	Class-weight	Dropout + class balance
Overfitting Risk	Low	Leveraged via early stopping
Training Time	Faster	Slower
Best Use Case	Robust Baseline	Allows complex fine-tuning

In summary, we see improved performance with SVM over MLP as shown by higher recall, precision metrics. These results suggest that the improved performance is a result of better modeling the linear nature of the data without an over-complicated model potentially overfitting the data.

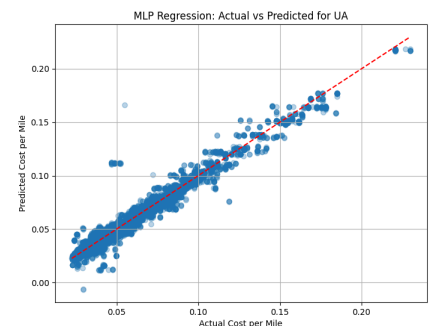
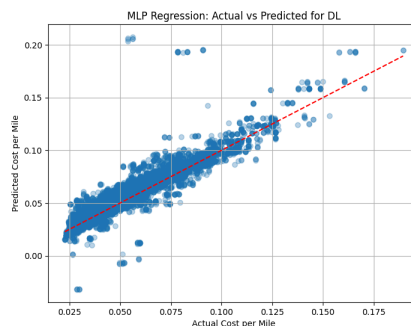
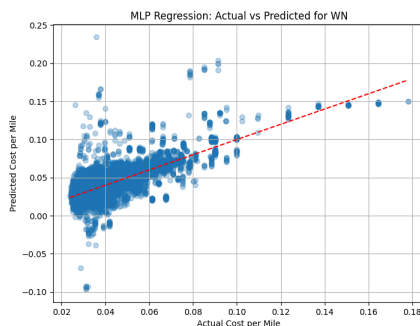
Cost per Mile Prediction:

To predict cost per mile, we implemented a regression model leveraging an MLP Regressor from the scikit-learn package. The model utilized features such as Distance, CRS Departure Time, Elapsed Time, and one-hot encoded locations. All features were standardized using the StandardScaler.

Our model achieved a Mean Absolute Error of approximately 0.018 and an R^2 score of around 0.87, indicating that it can explain about 87% of the variance in flight cost-per-mile. This performance is strong for an initial regression attempt. Below, we provide some computed metrics for the largest airlines. United Airlines demonstrates the best performance with an RMSE of 0.0178 and an R^2 score of 0.8912, which shows the model is both accurate and reliable. Delta Airlines ranks second best performance with an RMSE of 0.0186 and R^2 of 0.8879. Lastly, Southwest Airlines has the highest RMSE and lowest R^2 score of the three; therefore, since the performance is still good, the predictions are less precise and explain less variance compared with the other two airlines.

Airline	RMSE	R2 Score
Southwest	0.0191	0.8756
Delta	0.0186	0.8879
United	0.0178	0.8912

Actual vs. predicted (cost per mile) scatter plots for each airline (same three shown):



The plots above show the error for the cost per mile across the three airlines, with United Airlines (UA) demonstrating the lowest error. This is evident from the tightly clustered data points around the dashed line in the scatter plot and the low RMSE value, indicating that the predicted cost per mile aligns closely with the actual values. Delta Airlines (DL) shows slightly higher error levels compared to UA, as seen in a few data points deviating further from the ideal line. However, the model performs reliably for Delta, with only minor discrepancies. Southwest Airlines (WN), on the other hand, exhibits the largest errors among the three airlines. The wider spread of points around the dashed line and the higher RMSE value highlight the model's difficulty in accurately predicting the cost per mile for Southwest. This may reflect greater variability or noise in the WN dataset, leading to larger prediction errors. Overall, UA emerges as the best-performing airline in terms of prediction accuracy, followed by DL, while WN presents an opportunity for further model improvement.

Error Analysis

Delay prediction MLP implementation produced many erroneous predictions. The average accuracy was very low (60%), demonstrating that the model had difficulty making accurate predictions. On closer inspection, we see that the precision was extremely low (30%). This tells us that there were very few delayed flight predictions overall, meaning many delayed flights were classified incorrectly. Moreover, the recall was also low, demonstrating that many delayed flights were not caught. This may be because the dataset was on the smaller side, which is optimal for SVM, but a challenge for MLP, which may require much more data and features to generalize and is prone to overfitting. Additionally, the delayed flights show a linear relationship since it performs well with a linear SVM; therefore, the MLP non-linear implementation is not ideal for this dataset.

Regarding the Cost per Mile experiment, the limitation of our implementation stems from handling missing and approximated fields. Missing fields were either skipped or filled with averaged data from neighboring rows, depending on the implementation. Additionally, the metric was calculated using a proxy formula, which may have introduced noise. Refining this proxy formula could be an avenue for improvement. One of our MLP regressors demonstrated high training performance but low test R^2 , indicative of overfitting. The application of early stopping and dropout techniques improved generalization, as evidenced by smoothed loss curves.

Future Work

With additional time, we could improve the model in several ways. For hyperparameter tuning, dedicating more effort to the architecture of the MLP, such as adjusting layers, dropout rates, and activation functions, could enhance performance. Automated tuning methods like GridSearchCV would further streamline this process. In terms of accessibility, while the project already outputs plots and metrics per airline, developing a user-friendly web application would make the models and results more accessible to non-technical stakeholders, fostering practical applications of the insights. Addressing data challenges, such as refining the proxy formula for CostPerMile and exploring alternative methods for data imputation, could reduce noise and improve accuracy. Additionally, tackling the class imbalance, particularly for the “Severe” delay category, through advanced sampling techniques or synthetic data generation, could improve recall and overall model reliability. These steps would collectively enhance the model’s performance and usability.

Code

Public Repository: <https://github.com/TaisharTexas/JamScience>

References

Curran, Noah T., Thomas W. Kennings, and Kang G. Shin. "Analysis and Prevention of MCAS-Induced Crashes." IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS 43, no. 11 (November 2024).

Perkins, Johnny. "Competition in the Cockpit: US Air Travel amidst Global Disaster and Disruption." Harvard International Review, January 6, 2025.
<https://hir.harvard.edu/competition-in-the-cockpit-us-air-travel-amidst-global-disaster-and-disruption/>.

Shappell, S., Detwiler, C., Holcomb, K., Hackworth, C., Boquet, A., & Wiegmann, D. (2006). Human Error and Commercial Aviation Accidents: A Comprehensive, Fine-Grained Analysis Using HFACS. , (). Retrieved from <https://commons.erau.edu/publication/1218>

Singh, Shubham. "Flight Delay Dataset 2018-2024." Kaggle, June 23, 2024.
<https://www.kaggle.com/datasets/shubhamsingh42/flight-delay-dataset-2018-2024>